KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Year 10 Yearly Test Revision

Year 10 Transition · 2026 · Revision booklet, help sheet and answers

This booklet has practice questions for every type of question in the Year 10 Yearly Test. Write your answers in your **maths workbook** and show your working. A calculator is allowed.

Part 1 Revision questions: A Financial mathematics · B Measurement · C Statistics and probability · D Other Year 10 topics

Part 2 Help sheet: the formulas, rules and a worked example for every topic

Part 3 Answers with working, to check your work

Stuck? Every question has a tag like Help sheet A4. Turn to that box in Part 2 for the formula and an example, then try again.

Part 1 · Section A: Financial mathematics

Test questions 1–11

1 *Multiple choice* — Help sheet A1–A4

Choose the correct answer: A, B, C or D.

a What is 18% of $65?

Ⓐ $11.70 Ⓑ $18.00 Ⓒ $53.30 Ⓓ $83.00

b A price tag says “$99 (GST inc.)”. How much GST is included?

Ⓐ $9.00 Ⓑ $9.90 Ⓒ $90.00 Ⓓ $108.90

c A bag costs $40. It is discounted by 15%. What is the new price?

Ⓐ $6.00 Ⓑ $25.00 Ⓒ $34.00 Ⓓ $46.00

d Find the simple interest on $500 at 6% per year for 3 years.

Ⓐ $9.00 Ⓑ $30.00 Ⓒ $90.00 Ⓓ $590.00

2 *Percentage of an amount* — Help sheet A1

Find:

a 15% of $60 **b** 32% of $75 **c** 8% of $240 **d** 45% of $120

3 *GST* — Help sheet A2

a A price tag says “$66 (GST inc.)”. How much GST is included?

b A price tag says “$143 (GST inc.)”. How much GST is included?

c A haircut costs $80 before GST. Add 10% GST to find the price.

d A cap costs $77 including GST. What was the price before GST?

4 *Discounts* — Help sheet A3

Find the new (sale) price of each item.

a A $80 jacket with 25% off **b** A $45 game with 10% off **c** A $120 speaker with 35% off

5 *Simple interest* — Help sheet A4

a Find the simple interest on $800 at 5% per year for 3 years.

b Find the simple interest on $1500 at 4% per year for 3 years.

c Find the simple interest on $2000 at 3% per year for 5 years.

d For part c, what is the total value of the investment after 5 years?

6 *Percentage increase and decrease* — Help sheet A5

Use a multiplier and your calculator.

a Increase $250 by 8%

b Decrease $640 by 15%

c Increase $1200 by 12%

d Decrease $86 by 30%

7 *Mark-up and profit* — Help sheet A6

a A shop buys a sandwich press for $12 (the cost price) and adds a 40% mark-up.
i Find the selling price. ii How much profit does the shop make?

b A shop buys a phone case for $8.50 (the cost price) and adds a 60% mark-up.
i Find the selling price. ii How much profit does the shop make?

8 *Wages and overtime* — Help sheet A7

Kayla is paid $18.50 per hour at a café.

a One week she works 25 hours at the normal rate. How much is she paid?

b She also works 3 hours of overtime at time and a half. How much is she paid for the overtime?

c On Sunday she works 2 hours at double time. How much is she paid for Sunday?

d What is her total pay for the week?

9 *Commission* — Help sheet A8

How much commission does each person earn?

a Jai sells cars and earns 5% commission. His sales this week are $2400.

b Lena sells shoes and earns 3.5% commission. Her sales today are $860.

10 *Gross pay and net pay* — Help sheet A9

Sam works 38 hours at $24.50 per hour.

a Find Sam's gross pay (total pay before anything is taken out).

b Sam pays $160.40 in tax and $45 in super. Find his net pay (the pay he takes home).

11 *Loans and payment plans* — Help sheet A10

For each loan, find i the total amount repaid ii the interest paid.

a Ari buys a TV on a payment plan. He borrows $900 and repays $45 a month for 2 years.

b Bella borrows $2000 for a scooter. She repays $100 a month for 2 years.

12 *Compound interest* — Help sheet A11

For each investment, find i the value after the time given ii the interest earned.

a $3000 is invested for 2 years at 5% compound interest each year.

b $4000 is invested for 3 years at 3% compound interest each year.

Part 1 · Section B: Measurement

Test questions 12–21

13 *Multiple choice* — Help sheet B1–B4

Choose the correct answer: A, B, C or D.

a How many centimetres are in 1 metre?
Ⓐ 10 Ⓑ 100 Ⓒ 1000 Ⓓ 10 000

b What is the area of a rectangle 5 cm long and 2 cm wide?
Ⓐ 7 cm² Ⓑ 10 cm Ⓒ 14 cm Ⓓ 10 cm²

c What is the volume of a cube with sides 4 cm long?
Ⓐ 16 cm³ Ⓑ 12 cm³ Ⓒ 64 cm² Ⓓ 64 cm³

d Which of these is 56 000 written in scientific notation?
Ⓐ 5.6×10^4 Ⓑ 56×10^3 Ⓒ 5.6×10^{-4} Ⓓ 0.56×10^5

14 *Converting units* — Help sheet B1

Convert each measurement.

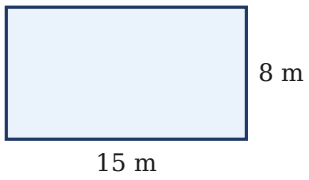
a 2.7 km into metres
b 4.2 L into millilitres
c 3500 m into kilometres
d 1500 mL into litres
e 6 m into centimetres
f 450 cm into metres

15 *Scientific notation* — Help sheet B2

a Write 45 000 in scientific notation.
b Write 3 200 000 in scientific notation.
c Write 670 000 in scientific notation.
d Write 4.1×10^3 as an ordinary number.

16 *Perimeter* — Help sheet B3

a A farmer puts a fence around this rectangular paddock. How many metres of fence are needed?



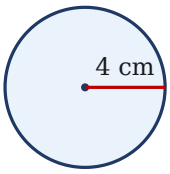
b Find the perimeter of a square with sides 6 cm long.

c A triangle has sides 5 m, 7 m and 9 m. Find its perimeter.

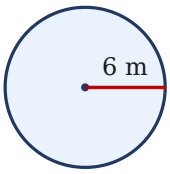
17 *Circles: circumference and area* — Help sheet B4

For each circle, find i the circumference ii the area. Give your answers to 1 decimal place.

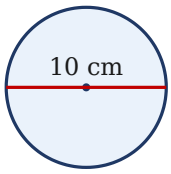
a



b



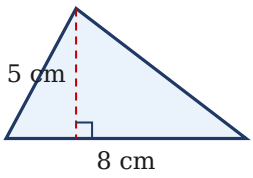
c



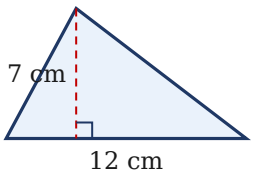
18 *Area of triangles and trapeziums* — Help sheet B5

Find the area of each shape.

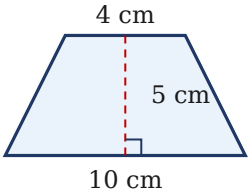
a



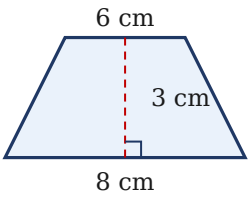
b



c



d

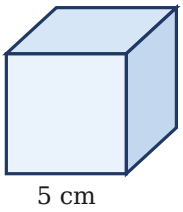


19 *Volume of cubes and rectangular prisms* — Help sheet B6

a Find the volume of a cube with sides 5 cm long.

b Find the volume of a box 6 cm long, 3 cm wide and 2 cm high.

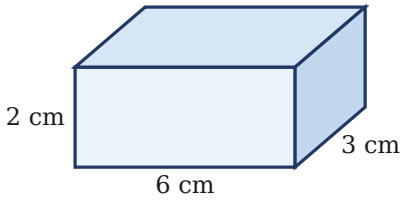
c A fish tank is 50 cm long, 20 cm wide and 30 cm high. Find its volume.



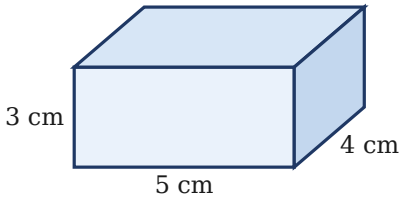
20 *Surface area of a box* Help sheet B7

Find the surface area of each box. Copy and complete a table like the one below for each box.

a



b

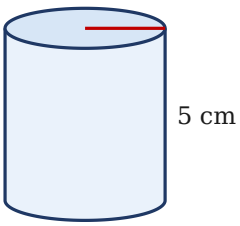


Faces	Area of one face	Number of faces	Area
Top and bottom		2	
Front and back		2	
The two ends		2	
Surface area (add the last column)			

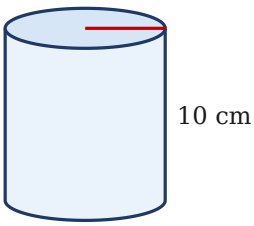
21 *Volume of a cylinder* Help sheet B8

For each cylinder, find i the volume, to 1 decimal place ii how many millilitres of water it can hold.

a



b



Part 1 · Section C: Statistics and probability

Test questions 22–29

22 *Multiple choice* Help sheet C1–C3

Choose the correct answer: A, B, C or D.

a A survey asks "What is your height in centimetres?" What type of data does this give?

Ⓐ numerical and discrete Ⓑ numerical and continuous
Ⓒ categorical and ordinal Ⓓ categorical and nominal

b A coin is tossed once. What is the probability of a tail?

Ⓐ 0 Ⓑ $\frac{1}{4}$ Ⓒ $\frac{1}{2}$ Ⓓ 1

23 *Types of data* Help sheet C1

Is each type of data **categorical** (nominal or ordinal) or **numerical** (discrete or continuous)?

a your favourite colour
b the number of pets you have
c your shirt size (S, M, L, XL)
d the time to run 100 m
e the number of goals in a soccer game
f your rating of a movie (poor, good, great)

24 *Stem-and-leaf plots* Help sheet C2

This stem-and-leaf plot shows the scores of 11 students in a quiz.

Stem	Leaf
1	5 8
2	3 3 3 7
3	0 4 6
4	1 5

3 | 4 means 34

a How many students scored in the 20s?
b What is the lowest score?
c What is the highest score?
d What is the mode (the most common score)?
e What is the range?
f What is the median (the middle score)?

25 *Mean, median, mode and range* — Help sheet C2

For each set of numbers, find i the mean ii the median iii the mode iv the range.

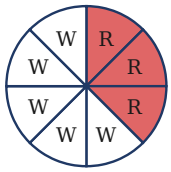
a The goals a hockey player scored in 5 games: 6, 2, 9, 4, 9

b The number of hours Ella spent on her phone over 7 days: 12, 7, 15, 10, 7, 4, 8

26 *Probability* — Help sheet C3

a A normal 6-sided dice is rolled once. Find the probability of rolling:
i a 5 ii an odd number iii a number greater than 4 iv a number less than 5

b This spinner has 8 equal parts. R is red and W is white. Find the probability of spinning:
i red ii white



27 *Probability with letters* — Help sheet C3

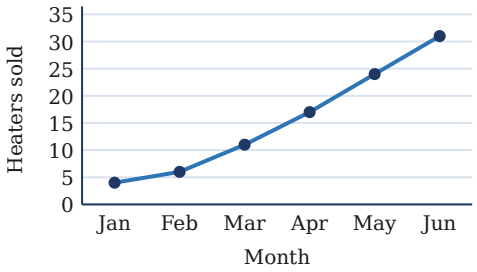
a One letter is chosen at random from the word **B A N A N A**.
Find the probability that the letter is: i an A ii an N iii a B

b One letter is chosen at random from the word **C O M M I T T E E**.
Find the probability that the letter is: i an M ii an E iii a vowel (A, E, I, O or U)

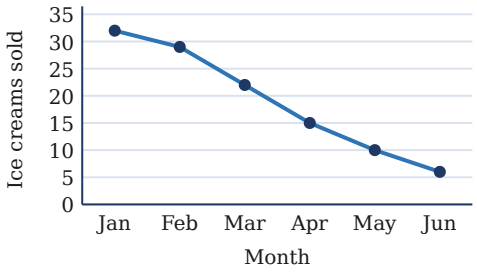
28 *Time-series graphs* — Help sheet C4

Describe the trend in each graph: **upward trend**, **downward trend** or **no trend**.

a



b



c Why do you think more heaters are sold in June than in January?

29 *Frequency tables and column graphs* — Help sheet C5

Some students recorded how many text messages they sent in one day.

Messages	0–9	10–19	20–29	30–39	40–49	Total
Frequency	4	8	5	2	1	

a Find the total number of students.

b Draw a column graph of the data in your workbook. Label both axes.

c What percentage of the students sent 10–19 messages?

d What percentage of the students sent 0–9 messages?

e Is the data symmetrical or skewed?

Part 1 · Section D: Other Year 10 topics

Test questions 30–35

30 *Multiple choice* — Help sheet D1

Choose the correct answer: A, B, C or D.

a Simplify $y^2 \times y^6$

Ⓐ y^8 Ⓑ y^{12} Ⓒ $2y^8$ Ⓓ $8y$

31 *Index laws* — Help sheet D1

Simplify each expression. Write your answer as a single power.

a $a^2 \times a^5$ **b** $m^3 \times m^6$ **c** $x \times x^4$

d $k^7 \times k^2$ **e** $b^8 \div b^3$ **f** $p^9 \div p^4$

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32 *Substitution* — Help sheet D2

Find the value of each expression.

a If $a = 3$, find $4a + 1$ **b** If $b = 5$, find $3b - 4$ **c** If $x = 2$, find $10 - 2x$

d If $y = 7$, find $20 - y$ **e** If $p = 6$, find $2p + 3$ **f** If $t = 4$, find $5t - 2$

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33 *Solving equations* — Help sheet D3

Solve each equation.

a $3x + 4 = 19$ **b** $5x - 2 = 18$ **c** $2x + 9 = 21$

d $\frac{x+3}{2} = 5$ **e** $\frac{x-1}{4} = 3$ **f** $\frac{x+5}{3} = 4$

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34 *Straight-line graphs* — Help sheet D4

a The rule for a straight line is $y = 3x + 1$. Copy and complete the table.

x	0	1	2	3
y				

b Plot the points from part a on a grid and join them with a straight line.

c Write the gradient and the y-intercept of $y = 3x + 1$.

d Write the gradient and the y-intercept of $y = x + 2$.

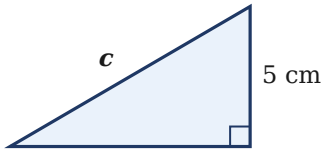
e Write the gradient and the y-intercept of $y = 4x + 5$.

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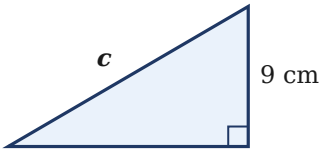
35 *Pythagoras' theorem* — Help sheet D5

Find the length of the hypotenuse, c. Give the answer to 1 decimal place where needed.

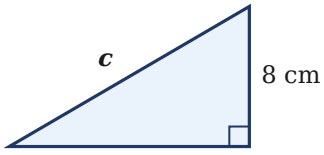
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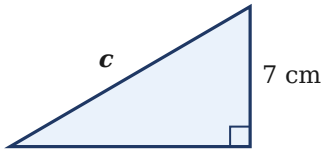
b



c



d



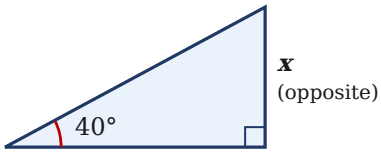
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36 *Trigonometry: finding a side*

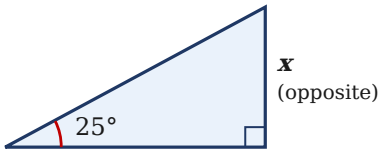
Help sheet D6

Find the length of side x, to 1 decimal place. Check your calculator is in degrees.

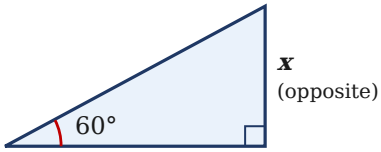
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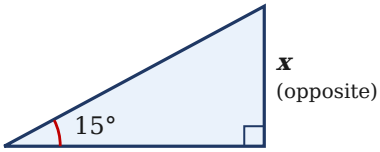
b



c



d



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Part 2 · Help sheet

Key ideas, formulas and worked examples for every Year 10 Yearly Test topic

A · Financial mathematics

A1 Percentage of an amount

Change the % into a decimal (divide by 100), then multiply.

> **Example:** 28% of \$50 → 0.28 × 50 = **\$14**

A2 GST (10%)

- Add GST: price × 1.1
- GST in a price that already has GST (GST inc.): price ÷ 11
- Price before GST: price ÷ 1.1

> **Example:** \$55 (GST inc.): GST = 55 ÷ 11 = **\$5**

A3 Discounts

Discount = price × rate. New price = price − discount.

> **Example:** \$60 with 20% off: 0.2 × 60 = \$12, 60 − 12 = **\$48**

A4 Simple interest

Formula
$I = P \times R \times N$

P = amount invested or borrowed, R = rate as a decimal (5% → 0.05), N = number of years. Total value = $P + I$.

> **Example:** \$400 at 5% for 2 years: 400 × 0.05 × 2 = **\$40**

A5 Increase or decrease with a multiplier

- Increase by 8% → × 1.08
- Decrease by 8% → × 0.92 (100% − 8% = 92%)

> **Example:** Increase \$50 by 10%: 50 × 1.10 = **\$55**

A6 Mark-up and profit

- Cost price = what the shop pays. Selling price = what the customer pays.
- Mark-up of 30% → cost × 1.3
- Profit = selling price − cost price

> **Example:** Cost \$10, mark-up 20%: 10 × 1.2 = \$12, profit = 12 − 10 = **\$2**

A7 Wages and overtime

- Pay = hours × hourly rate
- Time and a half = 1.5 × hourly rate
- Double time = 2 × hourly rate

> **Example:** \$20 per hour, 3 hours at time and a half: 3 × 1.5 × 20 = **\$90**

A8 Commission

Commission = rate (as a decimal) × sales.

> **Example:** 4% of \$900 sales: 0.04 × 900 = **\$36**

A9 Gross pay and net pay

- **Gross pay** = total pay before anything is taken out (hours × rate)
- **Net pay** = gross pay − tax − super (what you take home)

> **Example:** Gross \$800, tax \$120, super \$40: 800 − 120 − 40 = **\$640**

A10 Loans and payment plans

- Change years into months: 1 year = 12 months
- Total repaid = repayment × number of months
- Interest = total repaid − amount borrowed

> **Example:** Borrow \$500, repay \$50 a month for 1 year: 50 × 12 = \$600, interest = 600 − 500 = **\$100**

A11 Compound interest

Formula
$A = P(1 + R)^n$ A = final value, P = amount invested, R = rate as a decimal, n = number of years

> **Example:** \$2000 at 5% for 2 years: $A = 2000 \times 1.05^2$ = **\$2205**. Interest = 2205 − 2000 = **\$205**

> **Watch out:** 5% compound interest means × 1.05 each year. Calculator: type 2000 × 1.05 ^ 2 =. Interest = final value − amount invested.

B · Measurement

B1 Converting units

Length, capacity and mass
1 cm = 10 mm 1 m = 100 cm 1 km = 1000 m
1 L = 1000 mL 1 kL = 1000 L 1 kg = 1000 g

Big unit → small unit: **multiply**. Small unit → big unit: **divide**.

> **Example:** 5.3 km → m: 5.3 × 1000 = **5300 m**.
> 2500 mL → L: 2500 ÷ 1000 = **2.5 L**

B2 Scientific notation

A number between 1 and 10, times a power of 10. Count how many places the decimal point moves.

> **Example:** 2 350 000 = **2.35×10^6** (the point moves 6 places)

> **Example:** $4.1 \times 10^3 = 4.1 \times 1000 =$ **4100**

B3 Perimeter

Perimeter = add up the lengths of all the sides (the distance around the outside). Units: m, cm.

> **Example:** Rectangle 12 m by 7 m: 12 + 7 + 12 + 7 = **38 m**

B4 Circles

Formulas
Circumference $C = 2 \times \pi \times r$ Area $A = \pi \times r \times r$

r = radius (centre to edge). If you are given the diameter, halve it first.

> **Example:** Radius 5 cm: C = 2 × π × 5 = **31.4 cm**, A = π × 5 × 5 = **78.5 cm²**

B5 Area

Shape	Area
Rectangle	length × width
Triangle	½ × base × height
Trapezium	½ × (top + bottom) × height

Top and bottom are the parallel sides. The height makes a right angle with the base. Area units are squared: cm², m².

> **Example:** Triangle, base 8 cm, height 5 cm: ½ × 8 × 5 = **20 cm²**

B6 Volume of cubes and boxes

Formulas
Cube: V = side × side × side
Box (rectangular prism): V = length × width × height

Volume units are cubed: cm³, m³.

> **Example:** Box 4 cm × 3 cm × 2 cm: 4 × 3 × 2 = **24 cm³**

B7 Surface area of a box

A box has 6 faces in 3 matching pairs: top and bottom, front and back, the two ends. Find the area of one face of each pair, double it, then add.

> **Example:** Box 10 × 4 × 2: 2 × 40 + 2 × 20 + 2 × 8 = 80 + 40 + 16 = **136 cm²**

B8 Volume of a cylinder

Formula
$V = \pi \times r \times r \times h$

1 cm³ holds 1 mL, so the volume in cm³ is the capacity in mL.

> **Example:** r = 3 cm, h = 10 cm: π × 3 × 3 × 10 = **282.7 cm³** = 282.7 mL

Watch out: Round to 1 decimal place only at the end. Always write the units: cm for length, cm² for area, cm³ for volume.

C · Statistics and probability

C1 Types of data

Categorical (words or groups)	
Nominal	no order: eye colour, type of pet
Ordinal	has an order: small, medium, large
Numerical (numbers)	
Discrete	counted: number of goals
Continuous	measured: height, time, mass

C2 Mean, median, mode and range

- **Mean** = add up the numbers, then divide by how many numbers there are
- **Median** = the middle number when they are in order
- **Mode** = the most common number
- **Range** = biggest − smallest

> **Example:** 3, 8, 1, 8, 5 → in order 1, 3, 5, 8, 8. Mean = 25 ÷ 5 = **5**, median = **5**, mode = **8**, range = 8 − 1 = **7**

C2 Reading a stem-and-leaf plot

The stem is the tens digit. The leaf is the units digit. The key tells you how to read it.

Stem	Leaf
2	4 9
3	1 1 7

3 | 1 means 31

This plot shows 24, 29, 31, 31, 37. Lowest = 24, highest = 37, mode = 31.

C3 Probability

Formula

Probability = $\frac{\text{number of ways it can happen}}{\text{total number of ways}}$

Write the answer as a fraction and simplify it if you can.

> **Example:** A letter from BOOK: P(O) = $\frac{2}{4} = \frac{1}{2}$.
> Dice, an even number (2, 4, 6): $\frac{3}{6} = \frac{1}{2}$

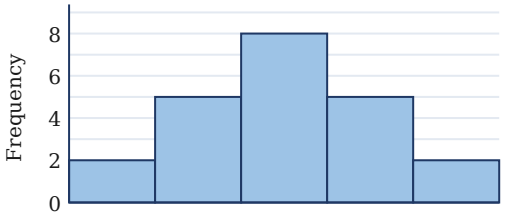
C4 Time-series graphs

- **Upward trend**: the line goes up over time
- **Downward trend**: the line goes down over time
- **No trend**: the line goes up and down with no pattern

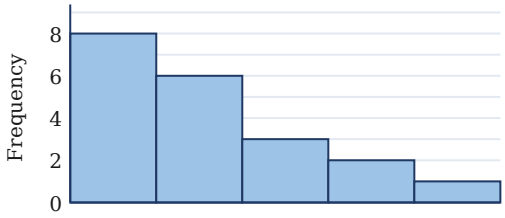
C5 Frequency tables, graphs and shape

- Total = add up the frequency column
- Percentage = frequency ÷ total × 100
- Column graph: frequency up the side, groups along the bottom
- **Symmetrical**: the same shape on both sides of the middle. **Skewed**: bunched at one end with a long tail

> **Example:** 9 out of 20 students: 9 ÷ 20 × 100 = **45%**



Symmetrical



Skewed

D · Other Year 10 topics

D1 Index laws

- When you **multiply**, add the powers: $m^2 \times m^6 = m^8$
- When you **divide**, subtract the powers: $m^7 \div m^2 = m^5$
- x on its own means x^1

D2 Substitution

Swap the letter for its number. $5a$ means $5 \times a$. Multiply before you add or subtract.

> **Example:** If $a = 3$: $2a + 1 = 2 \times 3 + 1 =$ **7**

D3 Solving equations

Do the opposite to both sides, one step at a time.

> **Example:** $3x + 1 = 13$ → take away 1: $3x = 12$ → divide by 3: $\boldsymbol{x = 4}$

> **Example:** $\frac{x + 1}{2} = 4$ → times by 2: $x + 1 = 8$ → take away 1: $\boldsymbol{x = 7}$

Check: put your answer back into the equation.

D4 Straight-line graphs

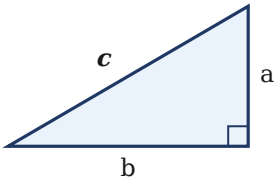
y* = *mx* + *c
m is the gradient (how steep the line is). *c* is the *y*-intercept (where the line crosses the *y*-axis).

Example: For $y = 3x + 2$, when $x = 4$: $y = 3 \times 4 + 2 = \mathbf{14}$. Gradient = 3, *y*-intercept = 2

To plot (4, 9): go across to 4, then up to 9. Put a dot. Join the dots with a ruler.

D5 Pythagoras' theorem

Formula
$c^2 = a^2 + b^2$ (*c* is the hypotenuse, the longest side, opposite the right angle)



Example: $a = 3$, $b = 4$: $c^2 = 9 + 16 = 25$, $c = \sqrt{25} = \mathbf{5}$

D6 Trigonometry: tan

Formula
tan (angle) = opposite ÷ adjacent
so opposite = adjacent × tan (angle)

Example: Angle 20°, adjacent 5 cm: $x = 5 \times \tan 20° = \mathbf{1.8\ cm}$

Watch out: The opposite side is across from the angle. The adjacent side is next to the angle (not the hypotenuse). Check your calculator shows D or DEG.

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Part 3 · Answers

Worked answers for the revision questions in Part 1

Check each answer. If yours is different, compare your working line by line to find where it went wrong. Answers are rounded at the last step.

Section A: Financial mathematics

1 *Multiple choice*

a 0.18 × 65 = \$11.70 → **A**

b 99 ÷ 11 = \$9 → **A**

c 40 × 0.15 = \$6, 40 − 6 = \$34 → **C**

d 500 × 0.06 × 3 = \$90 → **C**

2 *Percentage of an amount*

a 0.15 × 60 → **\$9**

b 0.32 × 75 → **\$24**

c 0.08 × 240 → **\$19.20**

d 0.45 × 120 → **\$54**

3 *GST*

a 66 ÷ 11 → **\$6**

b 143 ÷ 11 → **\$13**

c 80 × 1.1 → **\$88**

d 77 ÷ 1.1 → **\$70**

4 *Discounts*

a 0.25 × 80 = \$20, 80 − 20 → **\$60**

b 0.1 × 45 = \$4.50, 45 − 4.5 → **\$40.50**

c 0.35 × 120 = \$42, 120 − 42 → **\$78**

5 *Simple interest*

a I = 800 × 0.05 × 3 → **\$120**

b I = 1500 × 0.04 × 3 → **\$180**

c I = 2000 × 0.03 × 5 → **\$300**

d 2000 + 300 → **\$2300**

6 *Percentage increase and decrease*

a 250 × 1.08 → **\$270**

b 640 × 0.85 → **\$544**

c 1200 × 1.12 → **\$1344**

d 86 × 0.70 → **\$60.20**

7 *Mark-up and profit*

a i 12.00 × 1.4 → **\$16.80**

a ii 16.80 − 12.00 → **\$4.80**

b i 8.50 × 1.6 → **\$13.60**

b ii 13.60 − 8.50 → **\$5.10**

8 *Wages and overtime*

a 25 × 18.50 → **\$462.50**

b 3 × 1.5 × 18.50 → **\$83.25**

c 2 × 2 × 18.50 → **\$74**

d 462.50 + 83.25 + 74.00 → **\$619.75**

9 *Commission*

a 0.05 × 2400 → **\$120**

b 0.035 × 860 → **\$30.10**

10 *Gross pay and net pay*

a 38 × 24.50 → **\$931**

b 931.00 − 160.40 − 45.00 → **\$725.60**

11 *Loans and payment plans*

a i 2 years = 24 months, 45 × 24 → **\$1080**

a ii 1080 − 900 → **\$180**

b i 2 years = 24 months, 100 × 24 → **\$2400**

b ii 2400 − 2000 → **\$400**

12 *Compound interest*

a i $A = 3000 \times 1.05^2 \rightarrow$ **$3307.50**

a ii $3307.50 - 3000 \rightarrow$ **$307.50**

b i $A = 4000 \times 1.03^3 \rightarrow$ **$4370.91**

b ii $4370.91 - 4000 \rightarrow$ **$370.91**

Section B: Measurement

13 *Multiple choice*

a $1\text{ m} = 100\text{ cm} \rightarrow$ **B**

b $5 \times 2 = 10$, area is in $\text{cm}^2 \rightarrow$ **D**

c $4 \times 4 \times 4 = 64\text{ cm}^3 \rightarrow$ **D**

d $5.6 \times 10\,000 = 56\,000 \rightarrow$ **A**

14 *Converting units*

a $2.7 \times 1000 \rightarrow$ **2700 m**

b $4.2 \times 1000 \rightarrow$ **4200 mL**

c $3500 \div 1000 \rightarrow$ **3.5 km**

d $1500 \div 1000 \rightarrow$ **1.5 L**

e $6 \times 100 \rightarrow$ **600 cm**

f $450 \div 100 \rightarrow$ **4.5 m**

15 *Scientific notation*

a **4.5×10^4**

b **3.2×10^6**

c **6.7×10^5**

d $4.1 \times 1000 \rightarrow$ **4100**

16 *Perimeter*

a $15 + 8 + 15 + 8 \rightarrow$ **46 m**

b $6 + 6 + 6 + 6 \rightarrow$ **24 cm**

c $5 + 7 + 9 \rightarrow$ **21 m**

17 *Circles*

a i $C = 2 \times \pi \times 4 \rightarrow$ **25.1 cm**

a ii $A = \pi \times 4 \times 4 \rightarrow$ **50.3 cm²**

b i $C = 2 \times \pi \times 6 \rightarrow$ **37.7 m**

b ii $A = \pi \times 6 \times 6 \rightarrow$ **113.1 m²**

c i $r = 10 \div 2 = 5$; $C = 2 \times \pi \times 5 \rightarrow$ **31.4 cm**

c ii $A = \pi \times 5 \times 5 \rightarrow$ **78.5 cm²**

18 *Area of triangles and trapeziums*

a $\frac{1}{2} \times 8 \times 5 \rightarrow$ **20 cm²**

b $\frac{1}{2} \times 12 \times 7 \rightarrow$ **42 cm²**

c $\frac{1}{2} \times (4 + 10) \times 5 \rightarrow$ **35 cm²**

d $\frac{1}{2} \times (6 + 8) \times 3 \rightarrow$ **21 cm²**

19 *Volume of prisms*

a $5 \times 5 \times 5 \rightarrow$ **125 cm³**

b $6 \times 3 \times 2 \rightarrow$ **36 cm³**

c $50 \times 20 \times 30 \rightarrow$ **30 000 cm³**

20 *Surface area of a box*

a top and bottom $6 \times 3 = 18$, $\times 2 = 36$; front and back $6 \times 2 = 12$, $\times 2 = 24$; ends $3 \times 2 = 6$, $\times 2 = 12$; $36 + 24 + 12 \rightarrow$ **72 cm²**

b top and bottom $5 \times 4 = 20$, $\times 2 = 40$; front and back $5 \times 3 = 15$, $\times 2 = 30$; ends $4 \times 3 = 12$, $\times 2 = 24$; $40 + 30 + 24 \rightarrow$ **94 cm²**

21 *Volume of a cylinder*

a i $V = \pi \times 3 \times 3 \times 5 \rightarrow$ **141.4 cm³**

a ii 1 cm^3 holds 1 mL $\rightarrow$ **141.4 mL**

b i $V = \pi \times 2 \times 2 \times 10 \rightarrow$ **125.7 cm³**

b ii 1 cm^3 holds 1 mL $\rightarrow$ **125.7 mL**

Section C: Statistics and probability

22 *Multiple choice*

a height is measured $\rightarrow$ **B**

b 1 tail out of 2 sides $\rightarrow$ **C**

23 *Types of data*

a **Categorical, nominal (words, no order)**

b **Numerical, discrete (counted)**

c **Categorical, ordinal (groups with an order)**

d **Numerical, continuous (measured)**

e **Numerical, discrete (counted)**

f **Categorical, ordinal (has an order)**

---

24 *Stem-and-leaf plots*

a stem 2 has 4 leaves → **4 students**

b **15**

c **45**

d 23 appears 3 times → **23**

e 45 − 15 → **30**

f the 6th score of 11 → **27**

---

25 *Mean, median, mode and range*

a **In order: 2, 4, 6, 9, 9**

a i 30 ÷ 5 → **6**

a ii middle number → **6**

a iii most common → **9**

a iv 9 − 2 → **7**

b **In order: 4, 7, 7, 8, 10, 12, 15**

b i 63 ÷ 7 → **9**

b ii middle number → **8**

b iii most common → **7**

b iv 15 − 4 → **11**

---

26 *Probability*

a i 1 way out of 6 → $\frac{1}{6}$

a ii 1, 3, 5 → $\frac{3}{6} = \frac{1}{2}$

a iii 5, 6 → $\frac{2}{6} = \frac{1}{3}$

a iv 1, 2, 3, 4 → $\frac{4}{6} = \frac{2}{3}$

b i 3 red out of 8 → $\frac{3}{8}$

b ii 5 white out of 8 → $\frac{5}{8}$

---

27 *Probability with letters*

a i 3 A's out of 6 letters → $\frac{3}{6} = \frac{1}{2}$

a ii 2 N's out of 6 → $\frac{2}{6} = \frac{1}{3}$

a iii 1 B out of 6 → $\frac{1}{6}$

b i 2 M's out of 9 letters → $\frac{2}{9}$

b ii 2 E's out of 9 → $\frac{2}{9}$

b iii O, I, E, E: 4 out of 9 → $\frac{4}{9}$

---

28 *Time-series graphs*

a the line goes up → **Upward trend**

b the line goes down → **Downward trend**

c **June is winter in Australia (cold), January is summer (hot).**

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29 *Frequency tables and column graphs*

a 4 + 8 + 5 + 2 + 1 → **20 students**

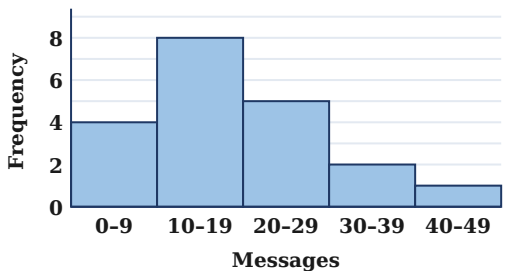
b **Columns of height 4, 8, 5, 2, 1 with no gaps; "Messages" on the horizontal axis, "Frequency" on the vertical axis.**

c 8 ÷ 20 × 100 → **40%**

d 4 ÷ 20 × 100 → **20%**

e most data is at the low end with a tail to the right → **Skewed**

Column graph for part b



Section D: Other Year 10 topics

30 *Multiple choice*

2 + 6 = 8 → **A**

31 *Index laws*

a add: 2 + 5 → a^7

b add: 3 + 6 → m^9

c add: 1 + 4 → x^5

d add: 7 + 2 → k^9

e subtract: 8 − 3 → b^5

f subtract: 9 − 4 → p^5

32 *Substitution*

a 4 × 3 + 1 → **13**

b 3 × 5 − 4 → **11**

c 10 − 2 × 2 → **6**

d 20 − 7 → **13**

e 2 × 6 + 3 → **15**

f 5 × 4 − 2 → **18**

33 *Solving equations*

a take away 4: 3x = 15; divide by 3 → **$x = 5$**

b add 2: 5x = 20; divide by 5 → **$x = 4$**

c take away 9: 2x = 12; divide by 2 → **$x = 6$**

d times by 2: $x + 3 = 10$; take away 3 → **$x = 7$**

e times by 4: $x - 1 = 12$; add 1 → **$x = 13$**

f times by 3: $x + 5 = 12$; take away 5 → **$x = 7$**

34 *Straight-line graphs*

a $y = 3 \times 0 + 1, 3 \times 1 + 1, \ldots$ → **$y = 1, 4, 7, 10$**

b **Points (0, 1), (1, 4), (2, 7), (3, 10) joined with a ruler.**

c **gradient = 3, y-intercept = 1**

d **gradient = 1, y-intercept = 2**

e **gradient = 4, y-intercept = 5**

35 *Pythagoras' theorem*

a $c^2 = 25 + 144 = 169$, $c = \sqrt{169}$ → **13 cm**

b $c^2 = 81 + 144 = 225$, $c = \sqrt{225}$ → **15 cm**

c $c^2 = 64 + 225 = 289$, $c = \sqrt{289}$ → **17 cm**

d $c^2 = 49 + 100 = 149$, $c = \sqrt{149}$ → **12.2 cm**

36 *Trigonometry*

a $x = 10 \times \tan 40°$ → **8.4 cm**

b $x = 6 \times \tan 25°$ → **2.8 cm**

c $x = 4 \times \tan 60°$ → **6.9 cm**

d $x = 20 \times \tan 15°$ → **5.4 cm**